Data Wrangling with Tidyverse

Data Science for the Psychological Sciences

Introduction: What is Data Wrangling?

Before you can build a predictive model, create an insightful plot, or find a meaningful pattern, you must first confront the raw material of your work: the data itself. Data in its original form is rarely, if ever, ready for analysis. It might be structured inconveniently, contain errors, or be missing critical information. The process of taking this raw, messy data and transforming it into a clean, organized, and usable format is known as **data wrangling** (or sometimes *data munging*).

Think of it like being a chef. You cannot prepare a gourmet meal until you've washed the vegetables, trimmed the fat, and organized your ingredients. Data wrangling is the essential preparation that makes all the exciting parts of data science possible.

The 80/20 Rule: The Importance of Data Preparation

There is a well-known adage that governs the daily life of nearly every data analyst, scientist, and researcher, often referred to as the "80/20 Rule of Data Science." The rule states that a data professional will spend approximately **80% of their time discovering, cleaning, and reorganizing data**, and only **20% of their time on the "core" tasks of analysis and modeling**. While this might seem surprising, it's a reality born from a simple truth: data in the wild is almost never ready for a statistical model or a visualization.

The 80% is the foundational, often unglamorous, work of data wrangling. It's the process of transforming raw data into a pristine, structured dataset. The 20% is the payoff—the part where you uncover insights and generate value. Without a solid foundation, any analysis you build will be unreliable at best and dangerously misleading at worst. This principle is famously captured in the phrase *"garbage in, garbage out."* The skills you learn in this chapter are designed to ensure you work with "gold in, gold out."

Raw data comes with a wide array of potential problems, including:

- **Missing values** that need to be handled or removed.
- **Inconsistent formatting**, such as dates written in different ways ("10-03-2025" vs. "Oct 3, 2025") or categorical data with typos ("Williamsburg", "williamsburg", "Wmsbg").
- **Incorrect data types**, like numbers stored as text that prevent mathematical operations.
- **A structure that is not "tidy"**, where data is organized for human readability (like a wide spreadsheet) but is unsuitable for R's analytical functions.

Mastering the tools to efficiently tackle these issues is what turns the frustrating 80% of your time into a powerful and controlled part of your workflow, paving the way for the insightful 20% that follows.

Chapter Goals

In this chapter, you will learn a complete data wrangling workflow using the core packages of the Tidyverse. By the end, you will be able to:

- Import data using `readr`.
- Wrangle missingness and messiness
- Select, filter, and create variables using `dplyr`.
- Reshape your data into a tidy format using `tidyr`.
- Summarize your data by groups.
- Create an insightful plot from your wrangled data.

Throughout this chapter, we will be working with a variety of datasets that are natively available in base R, exploring patterns and preparing the data for a hypothetical analysis.

Introduction to The Tidyverse

While base R has many powerful tools for data manipulation, the **Tidyverse** offers an ecosystem of packages designed specifically for the daily tasks of a data scientist. The packages share an underlying design philosophy and grammar, making the code you write more consistent, readable, and effective.

At the heart of this philosophy is the concept of stringing commands together using the **pipe operator** (`%>%` or `|>`), which allows you to read a sequence of data transformations like a sentence rather than a series of nested, hard-to-read functions.

The Principle of Tidy Data

The ultimate goal of data wrangling within the Tidyverse is to produce **tidy data**. A dataset is considered "tidy" when it conforms to three simple rules:

1. **Every column is a variable.**
2. **Every row is an observation.**
3. **Every cell contains a single value.**

Consider the difference. A "messy" dataset might be structured for easy data entry, like this:

City	Temp_2024	Temp_2025
Williamsburg	62	64
Richmond	61	63

This format is difficult for a program to analyze. The tidy version of this data would be:

City	Year	Temperature
Williamsburg	2024	62
Williamsburg	2025	64
Richmond	2024	61
Richmond	2025	63

This "long" format is far more flexible and is the standard that Tidyverse tools are designed to work with.

The Foundation: Pipes and Data Ingestion

Before we can start wrangling, we need two fundamental tools: a way to write clean, readable code and a way to reliably get data into R. In this section, we'll cover the pipe operator, which is the syntax that makes the Tidyverse so intuitive, and the `readr` package, our tool for importing data.

The Key to Readability: The Pipe Operator (%>% and |>)

Imagine you want to perform a series of operations on some data. In base R, you might end up nesting functions, like this: `function_c(function_b(function_a(data)))`

This code is difficult to read because you have to work from the inside out. It's also hard to debug; if something goes wrong, it's not immediately clear at which step the error occurred.

The Tidyverse solves this problem with the **pipe operator**. The most common pipe, from the `magrittr` package (which is included in the core `tidyverse`), is `%>%`.

The pipe takes the result of the expression on its left and passes it as the **first argument** to the function on its right. This allows you to rewrite the nested code above as a linear chain of operations:

```
data %>% function_a() %>% function_b() %>% function_c()
```

This is revolutionary for readability. You can now read your code from left to right, like a sentence: "Take the data, then do function A, then do function B, then do function C."

A Simple, Non-Data Example Let's see this in action with a simple character string. Suppose we want to take a messy string, convert it to lowercase, replace a word, and then remove leading/trailing whitespace.

The nested approach would be confusing:

```r
trimws(gsub("world", "R", tolower("  Hello World!  ")))
```

```
## [1] "hello R!"
```

The piped approach is clear and sequential:

```r
# The tidyverse package loads the pipe for us
library(tidyverse)

"  Hello World!  " %>%
  tolower() %>%
  gsub(pattern = "world", replacement = "R") %>%
  trimws()
```

```
## [1] "hello R!"
```

> **A Note on the Native R Pipe:** Since R version 4.1.0, a native pipe, `|>`, has been built into R. It functions very similarly for simple cases. Throughout this chapter, we will primarily use the Tidyverse pipe, `%>%`, as it is the historical standard for these packages and offers some additional features we may see later.

Importing Your Data with `readr`

Getting data into R is not a single function call; it is a short workflow: (1) identify the file type and structure, (2) read it with the right parser, (3) verify that the result matches expectations, and (4) make small, deliberate corrections (types, missing-value codes, column names) before doing analysis. This section emphasizes reliable defaults using the tidyverse, while showing where base R is sufficient.

The `readr` package, part of the core Tidyverse, provides a fast and friendly way to read rectangular data from delimited files.

For our examples to be self-contained and reproducible, we will first create our own sample comma-separated (CSV) file using the `writeLines()` function. In a real-world scenario, this file would already exist on your computer.

```
# Define the content for our example CSV file in a multi-line string
csv_content <- "year,visitors_millions,revenue_millions
2022,1.5,50.2
2023,1.7,55.8
2024,1.6,53.1"

# Write the content to a file named 'williamsburg_visitors.csv'
writeLines(csv_content, "williamsburg_visitors.csv")
```

Now that we have created the `williamsburg_visitors.csv` file, we can read it with a single function, `read_csv()`:

```
visitors <- read_csv("williamsburg_visitors.csv")
print(visitors)
```

```
## # A tibble: 3 x 3
##    year visitors_millions revenue_millions
##   <dbl>             <dbl>            <dbl>
## 1  2022               1.5             50.2
## 2  2023               1.7             55.8
## 3  2024               1.6             53.1
```

When we run `read_csv()`, it prints a helpful message listing the column specifications. It intelligently "guessed" that `year` is a double (a type of number), `visitors_millions` is a double, and `revenue_millions` is a double.

Note that `read_csv()` uses a number of named arguments with sensible defaults, such as `header=TRUE` and `sep=","`, which is why the function automatically parsed the first row of the file as column names and understood that each column was separated by a comma (`sep=","`), but the values of these arguments can be adjusted to meet the specific format of a particular data set. Other commonly used arguments include:

- `skip`: Skips a specified number of lines at the start of the file.
- `comment`: Specifies a character that indicates a line is a comment and should be ignored.

Later in this chapter, we will discuss issues related to missing data, sometimes called "missingness". At this point, it is simply important to recognize that many datasets encode missingness as strings like "NA", " ","N/A", "NULL", ".", or "-99". You should convert these to NA at import and this is easily accomplished

using the `na=` argument.

```
df <- read_csv(
  "data/survey.csv",
  na = c("", "NA", "N/A", "NULL", ".", "-99")
)
```

In fact, the `read_csv()` function is flexible enough to meet the needs of most data formats, however, other functions are available. For example…

- Tab-separated values (tsv) = `read_tsv()`
- Excel files = `read_excel("data/budget.xlsx", sheet = "FY2026", range = "A3:H250")`

Common Import Problems and Fixes

1. IDs and leading zeros

Sometimes researchers will use leading zeros in their ID column like `00123`. Because most import functions will interpret this column as "numeric", those values will automatically become `123`. If you need to keep the leading zeros, you can force character at import:

```
df <- read_csv("data/customers.csv", col_types = cols(customer_id = col_chara
```

2. Numbers with commas or currency symbols

Another common issue is data formatting that includes special characters like commas or currency symbols etc. These also can be dealt with by reading the data as character, then cleaning and converting them after import.

```
library(dplyr)
library(stringr)
library(readr)

df <- read_csv("data/sales.csv", col_types = cols(revenue = col_character()))
  mutate(revenue = parse_number(revenue))
```

`parse_number()` extracts the numeric portion of a character string and returns it as a numeric value. It is designed for messy "numbers in text," so it will correctly handle common formatting like currency symbols and thousands separators (e.g., "$1,234.50" becomes 1234.5), making it a reliable step when numeric columns were imported as character because of inconsistent formatting.

3. Dates in inconsistent formats

Date format inconsistencies can similarly be dealt with by importing as character first, then parsing deliberately.

```
df <- read_csv("data/orders.csv", col_types = cols(order_date = col_character
  mutate(order_date = as.Date(order_date, format = "%m/%d/%Y"))
```

Here `as.Date()` converts a value into an R Date object, which stores calendar dates (year–month–day) without any time-of-day component. It is commonly used after import to turn character strings into real

dates, and it is most reliable when you supply the format argument (e.g., format = "%m/%d/%Y") so R interprets the incoming text consistently rather than guessing.

Understanding the Output: Tibbles

The `readr` functions don't return a standard `data.frame`; they return a **tibble**. Tibbles are the Tidyverse's modern take on data frames. They work like data frames but are tweaked to be more user-friendly:

- **Printing:** They only print the first 10 rows and as many columns as fit on screen, preventing you from accidentally printing thousands of rows in your console. They also display the data type under each column name.
- **Consistency:** They are more consistent with subsetting. For example, using `[` to select columns will always return another tibble, whereas a data frame might sometimes return a vector.

Inspecting Your Import with `glimpse()`

A great way to get a quick overview of your newly imported data is with the `glimpse()` function from `dplyr`. It provides a transposed view of your data, making it easy to see every column, its data type, and the first few values.

```
glimpse(visitors)
```

```
## Rows: 3
## Columns: 3
## $ year              <dbl> 2022, 2023, 2024
## $ visitors_millions <dbl> 1.5, 1.7, 1.6
## $ revenue_millions  <dbl> 50.2, 55.8, 53.1
```

Cleaning the Column Names

One of the first issues that one notices when importing data is the column names. The "janitor" package provides small, high-impact utilities for cleaning common "messy data" issues. One of the most useful is `clean_names()`, which standardizes column names into a consistent, analysis-friendly format, typically lowercase snake_case, by removing punctuation, replacing spaces with underscores, and making names unique when duplicates exist. This is especially helpful when importing spreadsheets or CSVs with headers like "Order Date", "Total ($)", or "Customer ID", because consistent names reduce quoting/backticks and make pipelines easier to read and less error-prone.

```
library(janitor)
library(dplyr)

df <- df_raw %>%
  clean_names()  # e.g., "Order Date" -> "order_date", "Total ($)" -> "total"
```

Practical note: clean_names() changes the column names, so any downstream code should refer to the cleaned names (e.g., order_date rather than Order Date).

The Core Verbs of `dplyr`: Transforming Your Data

Now that we know how to get data into R, we can begin the real work of wrangling. Our primary tool for this is the `dplyr` package, the workhorse of the Tidyverse. The power of `dplyr` lies in its consistent and intuitive design. It provides a set of functions, or "verbs," where each verb corresponds to a common data manipulation task.

These verbs are designed to be chained together using the pipe operator (%>%), allowing you to write a clean, step-by-step recipe for transforming your data. For the examples in this section, we will use the `starwars` dataset, which is included with `dplyr`.

```
# Load the full tidyverse, which includes dplyr
library(tidyverse)

# Let's take a glimpse at our dataset
glimpse(starwars)
```

```
## Rows: 87
## Columns: 14
## $ name       <chr> "Luke Skywalker", "C-3PO", "R2-D2", "Darth Vader", "Leia
## $ height     <int> 172, 167, 96, 202, 150, 178, 165, 97, 183, 182, 188, 180
## $ mass       <dbl> 77.0, 75.0, 32.0, 136.0, 49.0, 120.0, 75.0, 32.0, 84.0,
## $ hair_color <chr> "blond", NA, NA, "none", "brown", "brown, grey", "brown"
## $ skin_color <chr> "fair", "gold", "white, blue", "white", "light", "light"
## $ eye_color  <chr> "blue", "yellow", "red", "yellow", "brown", "blue", "blu
## $ birth_year <dbl> 19.0, 112.0, 33.0, 41.9, 19.0, 52.0, 47.0, NA, 24.0, 57
## $ sex        <chr> "male", "none", "none", "male", "female", "male", "femal
## $ gender     <chr> "masculine", "masculine", "masculine", "masculine", "fem
## $ homeworld  <chr> "Tatooine", "Tatooine", "Naboo", "Tatooine", "Alderaan",
## $ species    <chr> "Human", "Droid", "Droid", "Human", "Human", "Human", "H
## $ films      <list> <"A New Hope", "The Empire Strikes Back", "Return of th
## $ vehicles   <list> <"Snowspeeder", "Imperial Speeder Bike">, <>, <>, <>, "
## $ starships  <list> <"X-wing", "Imperial shuttle">, <>, <>, "TIE Advanced x
```

As you can see, the `starwars` dataset contains information about characters from the Star Wars films, with a nice mix of character, numeric, and list columns.

Subsetting Your Data: `select()` and `filter()`

Two of the most fundamental data wrangling tasks are subsetting by columns and by rows.

Choosing Columns with `select()`

The `select()` verb is used to choose which columns you want to keep. You can specify columns by name.

```
# Select only the name, height, and mass columns
starwars %>%
  select(name, height, mass)
```

```
## # A tibble: 87 x 3
```

```
##    name                height  mass
##    <chr>                <int> <dbl>
##  1 Luke Skywalker         172    77
##  2 C-3PO                  167    75
##  3 R2-D2                   96    32
##  4 Darth Vader            202   136
##  5 Leia Organa            150    49
##  6 Owen Lars              178   120
##  7 Beru Whitesun Lars     165    75
##  8 R5-D4                   97    32
##  9 Biggs Darklighter      183    84
## 10 Obi-Wan Kenobi         182    77
## # i 77 more rows
```

You can also use a range of helper functions for more powerful selections: * Use `-` to exclude columns: `select(-films, -vehicles, -starships)` * Use a range with `:` `select(name:mass)` * Use helper functions like `starts_with()`, `ends_with()`, and `contains()`:

```r
# Select all columns that contain the word "color"
starwars %>%
  select(name, contains("color"))
```

```
## # A tibble: 87 x 4
##    name                hair_color    skin_color  eye_color
##    <chr>               <chr>         <chr>       <chr>
##  1 Luke Skywalker      blond         fair        blue
##  2 C-3PO               <NA>          gold        yellow
##  3 R2-D2               <NA>          white, blue red
##  4 Darth Vader         none          white       yellow
##  5 Leia Organa         brown         light       brown
##  6 Owen Lars           brown, grey   light       blue
##  7 Beru Whitesun Lars  brown         light       blue
##  8 R5-D4               <NA>          white, red  red
##  9 Biggs Darklighter   black         light       brown
## 10 Obi-Wan Kenobi      auburn, white fair        blue-gray
## # i 77 more rows
```

Choosing Rows with `filter()`

The `filter()` verb is used to choose which rows you want to keep based on logical conditions.

```r
# Find all droids
starwars %>%
  filter(species == "Droid")
```

```
## # A tibble: 6 x 14
##   name   height  mass hair_color skin_color  eye_color birth_year sex   ge
##   <chr>   <int> <dbl> <chr>      <chr>       <chr>           <dbl> <chr> <c
## 1 C-3PO     167    75 <NA>       gold        yellow            112 none  ma
```

```
## 2 R2-D2      96    32 <NA>       white, blue red                33 none  mas
## 3 R5-D4      97    32 <NA>       white, red  red                NA none  mas
## 4 IG-88     200   140 none       metal       red                15 none  mas
## 5 R4-P17     96    NA none       silver, red red, blue          NA none  fem
## 6 BB8        NA    NA none       none        black              NA none  mas
## # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
## #   vehicles <list>, starships <list>
```

```r
# Find all characters from the planet Tatooine with a height greater than 100
starwars %>%
  filter(homeworld == "Tatooine", height > 100)
```

```
## # A tibble: 9 x 14
##   name      height  mass hair_color skin_color eye_color birth_year sex   g
##   <chr>      <int> <dbl> <chr>      <chr>      <chr>          <dbl> <chr> <
## 1 Luke Sky~    172    77 blond      fair       blue            19   male  m
## 2 C-3PO        167    75 <NA>       gold       yellow         112   none  m
## 3 Darth Va~    202   136 none       white      yellow          41.9 male  m
## 4 Owen Lars    178   120 brown, gr~ light      blue            52   male  m
## 5 Beru Whi~    165    75 brown      light      blue            47   fema~ f
## 6 Biggs Da~    183    84 black      light      brown           24   male  m
## 7 Anakin S~    188    84 blond      fair       blue            41.9 male  m
## 8 Shmi Sky~    163    NA black      fair       brown           72   fema~ f
## 9 Cliegg L~    183    NA brown      fair       blue            82   male  m
## # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
## #   vehicles <list>, starships <list>
```

Inside `filter()`, you can use standard logical operators: * & or `,`: AND * `|`: OR * `!`: NOT * `%in%`: is in a set of values. This is a useful shortcut for multiple `|` conditions.

```r
# Find all characters whose eye color is blue or red
starwars %>%
  filter(eye_color %in% c("blue", "red"))
```

```
## # A tibble: 24 x 14
##    name      height  mass hair_color skin_color eye_color birth_year sex   g
##    <chr>      <int> <dbl> <chr>      <chr>      <chr>          <dbl> <chr> <
##  1 Luke Sk~    172    77 blond      fair       blue            19   male  m
##  2 R2-D2        96    32 <NA>       white, bl~ red             33   none  m
##  3 Owen La~    178   120 brown, gr~ light      blue            52   male  m
##  4 Beru Wh~    165    75 brown      light      blue            47   fema~ f
##  5 R5-D4        97    32 <NA>       white, red red             NA   none  m
##  6 Anakin ~    188    84 blond      fair       blue            41.9 male  m
##  7 Wilhuff~    180    NA auburn, g~ fair       blue            64   male  m
##  8 Chewbac~    228   112 brown      unknown    blue           200   male  m
##  9 Jek Ton~    180   110 brown      fair       blue            NA   <NA>  <
## 10 IG-88       200   140 none       metal      red             15   none  m
## # i 14 more rows
## # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
## #   vehicles <list>, starships <list>
```

Ordering and Finding Uniques: `arrange()` and `distinct()`

Sorting Rows with `arrange()`

The `arrange()` verb reorders the rows of your data frame based on the values in one or more columns. By default, it sorts in ascending order.

```
# Arrange characters from shortest to tallest
starwars %>%
  arrange(height)
```

```
## # A tibble: 87 x 14
##    name     height  mass hair_color skin_color eye_color birth_year sex   g
##    <chr>     <int> <dbl> <chr>      <chr>      <chr>          <dbl> <chr> <
##  1 Yoda         66    17 white      green      brown            896 male  n
##  2 Ratts T~     79    15 none       grey, blue unknown           NA male  n
##  3 Wicket ~     88    20 brown      brown      brown              8 male  n
##  4 Dud Bolt     94    45 none       blue, grey yellow            NA male  n
##  5 R2-D2        96    32 <NA>       white, bl~ red               33 none  n
##  6 R4-P17       96    NA none       silver, r~ red, blue         NA none  f
##  7 R5-D4        97    32 <NA>       white, red red               NA none  n
##  8 Sebulba     112    40 none       grey, red  orange            NA male  n
##  9 Gasgano     122    NA none       white, bl~ black             NA male  n
## 10 Watto       137    NA black      blue, grey yellow            NA male  n
## # i 77 more rows
## # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
## #   vehicles <list>, starships <list>
```

```
# To sort in descending order, use the desc() helper
starwars %>%
  arrange(desc(height))
```

```
## # A tibble: 87 x 14
##    name     height  mass hair_color skin_color eye_color birth_year sex   g
##    <chr>     <int> <dbl> <chr>      <chr>      <chr>          <dbl> <chr> <
##  1 Yarael ~    264    NA none       white      yellow          NA   male  n
##  2 Tarfful     234   136 brown      brown      blue            NA   male  n
##  3 Lama Su     229    88 none       grey       black           NA   male  n
##  4 Chewbac~    228   112 brown      unknown    blue           200   male  n
##  5 Roos Ta~    224    82 none       grey       orange          NA   male  n
##  6 Grievous    216   159 none       brown, wh~ green, y~       NA   male  n
##  7 Taun We     213    NA none       grey       black           NA   fema~ f
##  8 Rugor N~    206    NA none       green      orange          NA   male  n
##  9 Tion Me~    206    80 none       grey       black           NA   male  n
## 10 Darth V~    202   136 none       white      yellow          41.9 male  n
## # i 77 more rows
## # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
## #   vehicles <list>, starships <list>
```

```r
# You can sort by multiple columns to break ties
starwars %>%
  arrange(homeworld, name)
```

```
## # A tibble: 87 x 14
##    name     height  mass hair_color skin_color eye_color birth_year sex   g
##    <chr>     <int> <dbl> <chr>      <chr>      <chr>          <dbl> <chr> <
##  1 Bail Pr~    191    NA black      tan        brown             67 male  n
##  2 Leia Or~    150    49 brown      light      brown             19 fema~ f
##  3 Raymus ~    188    79 brown      light      brown             NA male  n
##  4 Ratts T~     79    15 none       grey, blue unknown           NA male  n
##  5 Lobot       175    79 none       light      blue              37 male  n
##  6 Jek Ton~    180   110 brown      fair       blue              NA <NA>  <
##  7 Nute Gu~    191    90 none       mottled g~ red               NA male  n
##  8 Ki-Adi-~    198    82 white      pale       yellow            92 male  n
##  9 Mas Ame~    196    NA none       blue       blue              NA male  n
## 10 Mon Mot~    150    NA auburn     fair       blue              48 fema~ f
## # i 77 more rows
## # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
## #   vehicles <list>, starships <list>
```

Finding Unique Values with `distinct()`

The `distinct()` verb is used to find unique values or unique combinations of values across columns.

```r
# Find all unique species in the dataset
starwars %>%
  distinct(species)
```

```
## # A tibble: 38 x 1
##    species
##    <chr>
##  1 Human
##  2 Droid
##  3 Wookiee
##  4 Rodian
##  5 Hutt
##  6 <NA>
##  7 Yoda's species
##  8 Trandoshan
##  9 Mon Calamari
## 10 Ewok
## # i 28 more rows
```

Computing/Recoding New Columns: `mutate()`

Perhaps the most powerful verb is `mutate()`, which allows you to add new columns or modify existing ones.

```r
# The 'height' is in cm. Let's create a new column with height in meters.
starwars %>%
  select(name, height, mass) %>%
  mutate(
    height_m = height / 100,
    bmi = mass / (height_m  2) # You can use a newly created column immediate
  )
```

```
## # A tibble: 87 x 5
##    name                height  mass height_m   bmi
##    <chr>                <int> <dbl>    <dbl> <dbl>
##  1 Luke Skywalker         172    77     1.72  26.0
##  2 C-3PO                  167    75     1.67  26.9
##  3 R2-D2                   96    32     0.96  34.7
##  4 Darth Vader            202   136     2.02  33.3
##  5 Leia Organa            150    49     1.5   21.8
##  6 Owen Lars              178   120     1.78  37.9
##  7 Beru Whitesun Lars     165    75     1.65  27.5
##  8 R5-D4                   97    32     0.97  34.0
##  9 Biggs Darklighter      183    84     1.83  25.1
## 10 Obi-Wan Kenobi         182    77     1.82  23.2
## # i 77 more rows
```

`mutate()` is often combined with the `if_else()` function to create conditional columns.

```r
# Create a category for characters based on their mass
starwars %>%
  select(name, mass) %>%
  mutate(
    mass_category = if_else(mass > 100, "Heavy", "Not Heavy")
  )
```

```
## # A tibble: 87 x 3
##    name                mass mass_category
##    <chr>              <dbl> <chr>
##  1 Luke Skywalker        77 Not Heavy
##  2 C-3PO                 75 Not Heavy
##  3 R2-D2                 32 Not Heavy
##  4 Darth Vader          136 Heavy
##  5 Leia Organa           49 Not Heavy
##  6 Owen Lars            120 Heavy
##  7 Beru Whitesun Lars    75 Not Heavy
##  8 R5-D4                 32 Not Heavy
##  9 Biggs Darklighter     84 Not Heavy
## 10 Obi-Wan Kenobi        77 Not Heavy
## # i 77 more rows
```

Other Column Management: `rename()`, `relocate()`, and `transmute()`

These are helpful "housekeeping" verbs for cleaning up your final dataset.

- `rename()`: Renames columns. The syntax is `rename(new_name = old_name)`.
- `relocate()`: Changes the order of columns.
- `transmute()`: A special version of `mutate()` that keeps *only* the columns you create.

```
starwars %>%
  # Rename the 'name' column to be more descriptive
  dplyr::rename(character = name) %>%
  # Create a new bmi column
  mutate(
    height_m = height / 100,
    bmi = mass / (height_m  2)
  ) %>%
  # Move the new bmi column to be right after mass
  relocate(bmi, .after = mass) %>%
  # Select a few columns to show the result
  select(character, mass, bmi, height_m)
```

```
## # A tibble: 87 x 4
##    character           mass   bmi height_m
##    <chr>              <dbl> <dbl>    <dbl>
##  1 Luke Skywalker        77  26.0     1.72
##  2 C-3PO                 75  26.9     1.67
##  3 R2-D2                 32  34.7     0.96
##  4 Darth Vader          136  33.3     2.02
##  5 Leia Organa           49  21.8     1.5
##  6 Owen Lars            120  37.9     1.78
##  7 Beru Whitesun Lars    75  27.5     1.65
##  8 R5-D4                 32  34.0     0.97
##  9 Biggs Darklighter     84  25.1     1.83
## 10 Obi-Wan Kenobi        77  23.2     1.82
## # i 77 more rows
```

And here is `transmute()` in action, which is useful when you only care about the new variables you've created.

```
# Calculate BMI but only keep the character name and the result
starwars %>%
  transmute(
    character = name,
    bmi = mass / ((height / 100)  2)
  )
```

```
## # A tibble: 87 x 2
##    character            bmi
##    <chr>              <dbl>
##  1 Luke Skywalker      26.0
##  2 C-3PO               26.9
```

```
##   3 R2-D2                34.7
##   4 Darth Vader          33.3
##   5 Leia Organa          21.8
##   6 Owen Lars            37.9
##   7 Beru Whitesun Lars   27.5
##   8 R5-D4                34.0
##   9 Biggs Darklighter    25.1
##  10 Obi-Wan Kenobi       23.2
## # i 77 more rows
```

By combining these core verbs, you can construct powerful and readable data wrangling pipelines to prepare any dataset for the next stages of analysis.

Aggregating Data: The Split-Apply-Combine Strategy

So far, we have learned how to transform our data on a row-by-row or column-by-column basis. We can select, filter, and create new variables. However, some of the most powerful insights in data analysis come from **aggregating** data—that is, collapsing many values down into a single summary statistic, like a count, an average, or a maximum.

To do this, we use a powerful mental model called the **Split-Apply-Combine** strategy:

1. **Split:** Break a large dataset into smaller groups based on the values of a categorical variable.
2. **Apply:** Apply a summary function (e.g., `mean()`, `sum()`) to each group independently.
3. **Combine:** Combine the results from each group into a new, smaller, and cleaner summary data frame.

This strategy is remarkably intuitive to implement in `dplyr` using just two key verbs: `group_by()` and `summarise()`.

Splitting Data into Groups with `group_by()`

The `group_by()` verb is the first step in our strategy. Its job is to take a data frame and add metadata to it that marks it as "grouped" by one or more columns. Visually, the data doesn't change much, but this grouping information fundamentally changes how subsequent `dplyr` verbs operate.

Let's group the `starwars` data by the `species` column.

```r
# We continue to use the tidyverse library and the starwars dataset
library(tidyverse)

starwars %>%
  group_by(species)
```

```
## # A tibble: 87 x 14
## # Groups:   species [38]
##    name     height  mass hair_color skin_color eye_color birth_year sex   g
##    <chr>     <int> <dbl> <chr>      <chr>      <chr>          <dbl> <chr> <
##  1 Luke Sk~    172    77 blond      fair       blue            19   male  m
##  2 C-3PO       167    75 <NA>       gold       yellow         112   none  m
##  3 R2-D2        96    32 <NA>       white, bl~ red             33   none  m
```

```
##   4 Darth V~    202   136 none       white      yellow          41.9 male  r
##   5 Leia Or~    150    49 brown      light      brown           19   fema~ f
##   6 Owen La~    178   120 brown, gr~ light      blue            52   male  r
##   7 Beru Wh~    165    75 brown      light      blue            47   fema~ f
##   8 R5-D4        97    32 <NA>       white, red red             NA   none  r
##   9 Biggs D~    183    84 black      light      brown           24   male  r
##  10 Obi-Wan~    182    77 auburn, w~ fair       blue-gray       57   male  r
## # i 77 more rows
## # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
## #   vehicles <list>, starships <list>
```

Notice the output. The data looks the same, but the header now says `Groups:   species [38]`. This tells us that any `dplyr` verb we apply next will be performed independently on each of the 38 species groups. You can also group by multiple variables, for example, `group_by(homeworld, species)`.

Applying Summaries with `summarise()`

Once the data is grouped, we can use `summarise()` (you may also see its alias, `summarize()`) to apply our summary functions. The `summarise()` verb collapses all the rows within each group into a single summary row.

Let's start with an ungrouped summary to see how it works on the entire dataset.

```
starwars %>%
  summarise(
    avg_height = mean(height, na.rm = TRUE),
    min_mass = min(mass, na.rm = TRUE),
    character_count = n()
  )
```

```
## # A tibble: 1 x 3
##   avg_height min_mass character_count
##        <dbl>    <dbl>           <int>
## 1       175.       15              87
```

Here, we created a new data frame with a single row containing three summary values. * We used `mean()` and `min()` to calculate statistics. The argument `na.rm = TRUE` is crucial; it tells R to ignore missing values (NA) when calculating the summary. Without it, any calculation involving an NA will result in NA. * We also used `n()`, a special `dplyr` function that counts the number of rows in the current group. Here, since there were no groups, it counted all the rows in the dataset.

Now for the magic: let's combine `group_by()` and `summarise()`.

```
# For each species, calculate the average mass and the number of characters.
species_summary <- starwars %>%
  group_by(species) %>%
  summarise(
    avg_mass = mean(mass, na.rm = TRUE),
    character_count = n()
  ) %>%
  arrange(desc(character_count))
```

```r
print(species_summary)
```

```
## # A tibble: 38 x 3
##    species  avg_mass character_count
##    <chr>       <dbl>           <int>
##  1 Human        81.3              35
##  2 Droid        69.8               6
##  3 <NA>         81                 4
##  4 Gungan       74                 3
##  5 Kaminoan     88                 2
##  6 Mirialan     53.1               2
##  7 Twi'lek      55                 2
##  8 Wookiee     124                 2
##  9 Zabrak       80                 2
## 10 Aleena       15                 1
## # i 28 more rows
```

This is the Split-Apply-Combine strategy in action. `dplyr` automatically: 1. **Split** the `starwars` data into groups by `species`. 2. **Applied** the `mean()` and `n()` functions to each group. 3. **Combined** the results into the new `species_summary` tibble.

We even piped the result into `arrange()` to immediately see the most numerous species!

The Importance of `ungroup()`

After you perform a grouped operation, the data frame often remains grouped. You can see this in the output of our `species_summary` if we just ran the `summarise` step—it would show `Groups: [species]`.

This can be problematic if you intend to perform further manipulations. For example, a subsequent `mutate()` would try to operate on a per-group basis, which might not be what you want.

It is considered good practice to explicitly remove the grouping information with `ungroup()` once you are finished with your summary calculations.

```r
# This pipeline is now "safe" because the final output is a regular, ungroupe
species_summary_safe <- starwars %>%
  group_by(species) %>%
  summarise(
    character_count = n(),
    avg_height = mean(height, na.rm = TRUE)
  ) %>%
  ungroup()

# Because we ungrouped, this mutate now works on the whole summary table,
# not on a per-species basis.
species_summary_safe %>%
  mutate(rank = min_rank(desc(character_count))) %>%
  arrange(rank)
```

```
## # A tibble: 38 x 4
##    species  character_count avg_height  rank
##    <chr>              <int>      <dbl> <int>
##  1 Human                 35       178      1
##  2 Droid                  6       131.     2
##  3 <NA>                   4       175      3
##  4 Gungan                 3       209.     4
##  5 Kaminoan               2       221      5
##  6 Mirialan               2       168      5
##  7 Twi'lek                2       179      5
##  8 Wookiee                2       231      5
##  9 Zabrak                 2       173      5
## 10 Aleena                 1        79     10
## # i 28 more rows
```

By adding `ungroup()` to the end of your aggregation pipelines, you ensure that your summary tables are clean, simple, and ready for the next steps of analysis or plotting without any surprising group-wise behavior.

Joins

Most real-world wrangling involves more than just a single table, it involves many tables. Joins are operations that combine two tables by matching their "keys". Join keys are one or more column names that identify the same entity in both datasets (for example, customer_id, date, or year). In a table, each row represents an observation (such as one customer, one transaction, or one day), and each column represents a variable or attribute (such as region, revenue, or temperature). A join works by taking the rows in one table and looking up corresponding rows in the other table where the key values match, then appending the other table's columns onto the result. The choice of join determines which rows are kept (e.g., all rows from the left table vs. only matched rows), while the quality of the result depends heavily on choosing keys that are consistent and appropriately unique for the level of detail in each table.

Overview of the Three Joins to Master

1. `left_join(x, y, by = ...)` = keep all rows from x

Use `left_join()` when x is your "main" table and y is supplemental. You keep every row in x; if a key is missing in y, new columns from y become NA.

- Best for: adding attributes (demographics, metadata, lookup tables)
- Risk: silently introduces missing values if keys don't match

2. `inner_join(x, y, by = ...)`: keep only matching rows

Use `inner_join()` when you want the intersection: rows that have keys present in both tables.

- Best for: restricting analysis to fully matched records
- Risk: silently drops records that don't match (can bias results if non-matching is systematic)

3. `anti_join(x, y, by = ...)`: find what does not match

`anti_join()` returns rows in x whose keys are not found in y. It is a diagnostic join: you use it to identify key mismatches, missing reference records, or unexpected values.

- Best for: validation and debugging
- Makes join problems visible immediately

Join keys: choosing by correctly

A join key is the variable (or set of variables) that identifies what "the same entity" means across tables: customer_id, date, station_id, year, etc.

- If the key has the same name, e.g., "date", in both tables, you can do:
  - `left_join(x, y, by = "date")`
- If the names differ between the two tables, map them explicitly using:
  - `left_join(x, y, by = c("date" = "obs_date"))`
  - where "date" is the join key in table x and "obs_date" is the join key in table y.

Composite keys are also common. A composite key means you need more than one column together to uniquely identify what a row represents. In many datasets, date alone is not unique. For example, you may have multiple rows per date (one per station, store, customer, product, etc.). Adding a second identifier (e.g., station_id) makes the key specific enough to match the correct records.

- If a composite key is required, use:
  - `left_join(x, y, by = c("date", "station_id"))`

Duplicates, many-to-many joins, and "join explosions"

The most common join failures are due to wrong cardinality. Cardinality describes the numeric relationship between two tables, namely, how many records in one side can be associated with a single record on the other side.

One-to-one: each key appears once in x and once in y

- Definition: each key appears once in x and once in y.

```r
library(tidyverse)
x <- tribble(
  ~id, ~x_val,
   1,  "A",
   2,  "B",
   3,  "C"
)

y <- tribble(
  ~id, ~y_val,
   1,  10,
   2,  20,
   3,  30
)

dplyr::left_join(x, y, by = "id")
```

```
## # A tibble: 3 x 3
##      id x_val y_val
```

```
##   <dbl> <chr> <dbl>
## 1     1 A        10
## 2     2 B        20
## 3     3 C        30
```

One-to-many Cardinality: a key appears once in x, multiple times in y

```r
x <- tribble(
  ~id, ~customer_name,
   1,  "Ivy",
   2,  "Max"
)

y <- tribble(
  ~id, ~order_id,
   1,  "o-100",
   1,  "o-101",
   2,  "o-200"
)

left_join(x, y, by = "id")
```

```
## # A tibble: 3 x 3
##      id customer_name order_id
##   <dbl> <chr>         <chr>
## 1     1 Ivy           o-100
## 2     1 Ivy           o-101
## 3     2 Max           o-200
```

Notice:

- id = 1 appears once in x but twice in y.
- Joining "replicates" the matching x row for each matching row in y.
- Row count can increase: 2 in → 3 out (because y has more matches per key).

This is not inherently wrong; it's correct if you intend to "attach events to entities." It becomes wrong when you expected one row per key after the join.

Many-to-one Cardinality: multiple in x, once in y

```r
x <- tribble(
  ~id, ~order_id,
   1,  "o-100",
   1,  "o-101",
   2,  "o-200"
)

y <- tribble(
  ~id, ~customer_name,
   1,  "Ivy",
   2,  "Max"
```

```r
)

left_join(x, y, by = "id")
```

```
## # A tibble: 3 x 3
##      id order_id customer_name
##   <dbl> <chr>    <chr>
## 1     1 o-100    Ivy
## 2     1 o-101    Ivy
## 3     2 o-200    Max
```

Notice: - x has multiple rows per customer; y has one row per customer. - Row count typically stays the same for a left join: 3 in → 3 out. - This is a "safe enrichment" pattern: you add attributes to many records.

Many-to-many Cardinality: multiple in both x and y (often accidental)

```r
x <- tribble(
  ~id, ~x_event,
   1,  "x1",
   1,  "x2",
   2,  "x3"
)

y <- tribble(
  ~id, ~y_event,
   1,  "y1",
   1,  "y2",
   2,  "y3"
)

left_join(x, y, by = "id")
```

```
## # A tibble: 5 x 3
##      id x_event y_event
##   <dbl> <chr>   <chr>
## 1     1 x1      y1
## 2     1 x1      y2
## 3     1 x2      y1
## 4     1 x2      y2
## 5     2 x3      y3
```

Notice:

- For id = 1, there are 2 rows in x and 2 rows in y.
- The join produces 2 × 2 = 4 rows for id = 1 (Cartesian pairing).
- Total row count increases unexpectedly: 3 in - 5 out here.

This is the easiest way to demonstrate the phrase "join explosion."

What is a join explosion?

If both tables have duplicates for the same key, joining produces a Cartesian product for that key. Example: if x has 3 rows for date = 2025-07-01 and y has 4 rows for the same date, the join produces 12 rows for that date. This can inflate totals, distort averages, and create subtle downstream errors.

Detect duplicates in keys before joining

One simple way to check for duplicates before joining is to count the unique values. For example, a simple pre-join check is:

```
x %>% count(id) %>% filter(n > 1)
```

```
## # A tibble: 1 x 2
##      id     n
##   <dbl> <int>
## 1     1     2
```

```
y %>% count(id) %>% filter(n > 1)
```

```
## # A tibble: 1 x 2
##      id     n
##   <dbl> <int>
## 1     1     2
```

If you expect a "one row per date" table, any n > 1 is a red flag.

Validating joins: row counts and "what didn't match"

A good join is one you can defend. A minimal validation checklist:

1. Row counts before/after

```
nrow(x)
nrow(y)

xy <- left_join(x, y, by = "date")
nrow(xy)
```

For a left join, nrow(xy) should be:

- equal to nrow(x) if keys in y are unique
- greater than nrow(x) if y has duplicates on the join key (possible join explosion)

2. Unmatched keys using `anti_join()`

```
x %>% anti_join(y, by = "date")   # dates present in x but missing in y
y %>% anti_join(x, by = "date")   # dates present in y but missing in x
```

Concise Join Example: Williamsburg visits + weather by date + demographics by year

This example mirrors a realistic "neighborhood story" dataset integration:

- visits_daily: daily counts of visits in Williamsburg (one row per day)

- weather_daily: daily weather (one row per day)
- demo_yearly: demographics by year (one row per year)

```r
library(dplyr)
library(lubridate)

# visits_daily: date, visits, avg_spend
# weather_daily: date, temp_max, precip_mm
# demo_yearly: year, population, median_income
visits_daily <- tribble(
  ~date,        ~visits, ~avg_spend,
  as.Date("2025-07-01"), 1200,   18.5,
  as.Date("2025-07-02"), 1350,   19.1,
  as.Date("2025-07-03"),  980,   17.8
)

weather_daily <- tribble(
  ~date,        ~temp_max, ~precip_mm,
  as.Date("2025-07-01"),  86,       0.0,
  as.Date("2025-07-02"),  88,       2.5,
  as.Date("2025-07-03"),  83,       0.0
)

demo_yearly <- tribble(
  ~year, ~population, ~median_income,
   2025,     152000,         84500
)

# 1) Join daily visits to daily weather (date-level integration)
visits_weather <- visits_daily %>%
  left_join(
    weather_daily,
    by = "date",
    relationship = "many-to-one"  # visits may have multiple rows per date; we
  )

# Validate: what dates in visits have no weather?
missing_weather_dates <- visits_daily %>%
  anti_join(weather_daily, by = "date")

# 2) Add demographics by year (derive year from date, then join)
visits_weather_demo <- visits_weather %>%
  mutate(year = year(date)) %>% #Make new "year" column for join
  left_join(demo_yearly, by = "year")

visits_weather_demo
```

```
## # A tibble: 3 x 8
```

```
##   date       visits avg_spend temp_max precip_mm  year population median_i
##   <date>      <dbl>     <dbl>    <dbl>     <dbl> <dbl>      <dbl>
## 1 2025-07-01   1200      18.5       86       0    2025     152000
## 2 2025-07-02   1350      19.1       88       2.5  2025     152000
## 3 2025-07-03    980      17.8       83       0    2025     152000
```

This example is meant to illustrate that joins often happen at different "grains" (daily vs yearly) and you sometimes have to derive a join key (year = year(date)) to align them.

Quick rules of thumb for learners

- Start with `left_join()`, validate with `anti_join()`, and use `inner_join()` only when you explicitly want to restrict to matched records.
- Always know the grain of each table (one row per what?) and ensure join keys reflect that grain.
- Treat unexpected row-count increases as a likely duplicate key problem until proven otherwise.
- Make join validation a habit: row counts, missingness in joined columns, and anti-joins are fast and revealing.

Dealing with Missing data

Earlier, we used the `na.rm = TRUE` argument inside `summarise()`, which is a useful tactical move in that it prevents missing values from breaking aggregations inside the `summarise()` function. But it does not solve the broader, more consequential questions:

1. Where is data missing, and how much?
2. Is missingness concentrated in certain variables, groups, or time periods?
3. Should we filter, impute, or explicitly model missingness?
4. What bias might we introduce by dropping rows or imputing?

This section introduces a practical "missing data toolkit" in tidyr (and complementary dplyr patterns) so learners can diagnose, decide, and act.

Core tidyr tools

There are two tidyr functions that are essential to effectively dealing with missingness:

1. `drop_na()`: filter rows with missing values

- drop_na() removes rows if any column contains a missing value.
- drop_na(customer_id, order_date) removes rows ONLY if customer_id and/or order_date columns contain a missing value.

The ability to focus on "key" variables is important because most analyses focus only on a subset of the critical variables (IDs, outcome variables, timestamps). Dropping rows based on all columns is often too aggressive.

2. `replace_na()`: fills missing values using fixed replacements (not model-based imputation). It is most appropriate when the replacement is meaningful (e.g., missing counts imply 0, missing categorical implies "Unknown"). For example:

```
df_filled <- df %>%
  replace_na(list(
```

```
    n_orders = 0,
    revenue  = 0,
    segment  = "Unknown"
  ))
```

It is important to note that `replace_na()` is a rule-based fill, not statistical imputation. If "0" is not a truthful value, this can distort results.

Missingness diagnostics

Before you start dropping or filling anything, it is good practice to quantify missingness. The goal is to produce a compact table showing:

- missing count (n_missing)
- missing percent (pct_missing)
- optionally: non-missing count

One way to achieve this simple missingness summary by column is illustrated below:

```
missing_by_col <- df %>%                                  #Assigns the final re
  summarise(across(everything(), ~ sum(is.na(.x)))) %>%
  pivot_longer(
    cols = everything(),
    names_to = "column",
    values_to = "n_missing"
  ) %>%
  mutate(
    pct_missing = n_missing / nrow(df)
  ) %>%
  arrange(desc(pct_missing))

missing_by_col
```

Let's break that down in order to understand each separate pipe.

1. `missing_by_col <- df %>%` = Assigns the name "missing_by_col" to our final output and starts by sending the data frame "df" into the first pipe.
2. `summarise(across(everything(), ~ sum(is.na(.x)))) %>%` = summarize all columns of df by summing the number of NAs.

- `across(everything(),function)` = applies a function across columns and `everything()` means all columns.
- `~ sum(is.na(.x)) = is the anonymous function applied to each column -.xis the current column vector. -is.na(.x)returns a logical vector: TRUE where values are missing and FALSE otherwise. -sum(is.na(.x))'` works because in R, TRUE is treated as 1 and FALSE as 0 in numeric contexts, so the sum is the count of missing values in that column.
- Note that `summarise()` returns one row per group, but because df is not grouped here, it returns exactly one row total. After this line, you have a one-row tibble with the same column names as df,

but each value is the missing count for that column

3. `pivot_longer(cols = everything(), names_to = "column", values_to = "n_missing")` %>% = This converts the one-row "wide" summary into a "long" two-column key/value structure (plus values).

- `pivot_longer(...)`: A tidyr function that takes multiple columns and turns them into rows.
- `cols = everything()`: Pivot all columns (which, at this stage, are the missing-count columns from the previous step).
- `names_to = "column"`: Put the original column names (e.g., colA, colB) into a new column called "column".
- `values_to = "n_missing"`: Put the numeric counts into a new column called "n_missing".

4. `mutate(pct_missing = n_missing / nrow(df))` %>% = mutate(...): Adds new columns (or transforms existing ones) without changing the number of rows.

- `pct_missing = n_missing / nrow(df)`:
  - `n_missing` is the count of missing values per column (from the previous step).
  - `nrow(df)` is the number of rows in the original dataset.
  - Dividing gives the proportion of rows that are missing for each column.

5. `arrange(desc(pct_missing))`

- `arrange(...)`: Sorts rows.
- `desc(pct_missing)`: Sort in descending order, so columns with the highest percent missing appear at the top.

To add non-missing counts, just add something like, n_nonmissing = nrow(df) - n_missing to the mutate pipe.

Missingness by subgroup (a fast way to detect non-random patterns)

Overall missingness can hide important structure. Missingness that differs by region, time, cohort, or treatment group is often a warning sign for bias.

Example: percent missing by column within a grouping variable:

```
missing_by_group <- df %>%
  group_by(region) %>%
  summarise(across(everything(), ~ mean(is.na(.x)))) %>%  # mean of TRUE/FALS
  pivot_longer(-region, names_to = "column", values_to = "pct_missing") %>%
  arrange(desc(pct_missing))

missing_by_group
```

Notice the only difference is that we used `group_by` in the second pipe, and used all columns EXCEPT "region" (`-region`) in the `pivot_longer` pipe.

Row-wise diagnostics: "How incomplete are my records?"

Sometimes the question is not "which columns are missing data?" but "which rows are unreliable?" A simple per-row missing count helps learners decide whether to drop highly incomplete records (i.e., "participants").

Below is an example of a row-wise missingness table:

```r
df_row_missing <- df %>%
  mutate(
    n_missing_row = rowSums(is.na(across(everything()))),
    pct_missing_row = n_missing_row / ncol(df)
  )

df_row_missing %>%
  count(n_missing_row) %>%
  arrange(desc(n_missing_row))

#Then apply a rule (example: drop rows missing more than 40% of fields):
df_filtered <- df_row_missing %>%
  filter(pct_missing_row <= 0.4) %>%
  select(-n_missing_row, -pct_missing_row)
```

Using these column-based and row-based diagnostics is important because missingness has the potential to introduce **bias**. For example, if missingness differs by subgroup (region, cohort, outcome class), then filtering can disproportionately remove specific populations and produce biased estimates.

String cleaning and categorical standardization

Real-world categorical fields are rarely analysis-ready. Variants like "NY", "New York", "new-york", and "N.Y." may represent the same concept but appear as distinct values, creating duplicated categories, misleading counts, and noisy plots. A robust wrangling workflow standardizes strings first (format and whitespace), then normalizes known variants into a canonical set of categories, and finally converts the result to factors for stable ordering and cleaner summaries.

A practical pattern: normalize → recode → validate

Of course, the first step is to inspect the problem. Start by identifying what's actually in the field.

```r
df %>%
  count(raw_category, sort = TRUE)
```

This quickly reveals the typos and near-duplicates you will want to address. The next step is to normalize the string data and (usually remove punctuation too)

Normalizing String Data

Use functions from the `stringr` package to apply consistent casing, trim whitespace, and remove or standardize punctuation.

```r
library(stringr)
library(dplyr)

df_clean <- df %>%
  mutate(
    category_norm = raw_category %>%
      str_to_lower() %>%                    # normalize case
      str_trim() %>%                        # remove leading/trailing whitespac
```

```
      str_replace_all("\\s+", " ") %>%    # collapse repeated internal spaces
      str_replace_all("[[:punct:]]+", " ")# replace punctuation with spaces
  )
```

What each function contributes

- `str_to_lower()`: prevents "Retail" and "retail" from being separate categories.
- `str_trim()`: handles trailing spaces that create invisible duplicates ("ny" vs "ny").
- `str_replace_all("\\s+", " ")`: collapses "double spaces" that survive imports or copy/paste.
- `str_replace_all("[[:punct:]]+", " ")`: converts punctuation into a uniform delimiter; helpful when variants include hyphens, periods, slashes, etc.

If punctuation replacement creates extra spaces, add another str_trim() at the end.

Now use `case_when()` to complete the normalization by mapping messy variants to canonical labels. This step is where domain knowledge enters.

```
df_std <- df_clean %>%
  mutate(
    category_std = case_when(
      category_norm %in% c("ny", "new york", "n y", "n y c", "nyc") ~ "New Yo
      category_norm %in% c("ca", "california")                      ~ "Califor
      category_norm %in% c("uk", "u k", "united kingdom", "britain") ~ "Unite
      is.na(category_norm) | category_norm == ""                    ~ NA_chara
      TRUE                                                          ~ str_to_t
    )
  )
```

What it does:

- `case_when()` is a readable "rulebook." It encourages documentation of decisions directly in code.
- Use `NA_character_` explicitly to keep case_when() type-stable (important in teaching and debugging).
- The final `TRUE ~ …` catch-all prevents accidental conversion to missing values and ensures unmatched categories remain visible for review.

Validate that standardization worked

Validation is not optional in practice. You want to confirm you reduced duplicates without silently creating new ones.

```
df_std %>%
  count(category_std, sort = TRUE)
```

You might also consider counting the distinct elements before/after

```
df %>% summarise(n_distinct_raw = n_distinct(raw_category))
df_std %>% summarise(n_distinct_std = n_distinct(category_std))
```

A useful diagnostic is to find categories that still look suspiciously similar (e.g., leftover variants). At a basic level, sorting by frequency highlights remaining "rare" categories that may be typos.

Factor handling for stable plots and summaries

Once categories are standardized, converting them to factors supports consistent ordering in plots and tables. This is particularly important when you want "Top categories first" or when you need a specific semantic order.

```
library(forcats)
library(dplyr)

df_fact <- df_std %>%
  mutate(
    category_f = factor(category_std),
    category_f = fct_relevel(category_f, "New York", "California", "United Ki
  )
```

By default, factor levels will be printed in ascending alphabetic order. `fct_relevel()` is used here for explicit ordering which is useful when you want a consistent plot order across all analyses.

What if you have a large number of categories/levels in a categorical variable? This is referred to as "high-cardinality". In order to improve the interpretability of some analysees and/or plots, it can be useful to lump the least frequent categories into an "other" category.

```
df_lumped <- df_std %>%
  mutate(
    category_f = factor(category_std),
    category_f = fct_lump(category_f, n = 10)  # keep top 10 categories, lump
  )
```

This is often preferable to manually filtering out rare categories because it keeps the full dataset while improving interpretability.

Reshaping Your Data with `tidyr`

We've now mastered the `dplyr` verbs for transforming the *values* within our data—selecting, filtering, arranging, and summarizing. But sometimes, the problem isn't with the values themselves, but with the *layout* of the data. This is where the `tidyr` package comes in. `tidyr` is designed to change the shape or layout of your data, helping you create the "tidy" data structure that is the ultimate goal of the wrangling process.

As a reminder, the three rules of tidy data are: 1. Every column is a variable. 2. Every row is an observation. 3. Every cell is a single value.

The two most important functions in `tidyr` for achieving this are `pivot_longer()` and `pivot_wider()`.

When Data Isn't Tidy: Wide vs. Long Formats

Data often comes in one of two formats: wide or long.

- **Wide Format:** This format is often found in spreadsheets and reports. It's easy for humans to scan across, but it violates the principles of tidy data. In wide data, you'll often find columns that are not

variables, but rather *values* of a variable (e.g., columns named `2023`, `2024`, `2025`). This format is difficult to use in R for plotting and modeling.

- **Long Format:** This is the tidy format. Every row is a single observation. This format is machine-readable and is what the Tidyverse is designed to work with. It's much easier to group, summarize, and plot data in a long format.

For example, imagine we have data on visitor numbers for historical sites in Williamsburg. The wide (messy) format might look like this:

site	2023	2024	2025
Governor's Palace	305k	312k	315k
Capitol Building	280k	295k	302k

The tidy (long) version of the same data would be:

site	year	visits
Governor's Palace	2023	305k
Governor's Palace	2024	312k
...	...	...

From Wide to Long: `pivot_longer()`

`pivot_longer()` is the workhorse function you will use most often. It takes multiple columns and "lengthens" them by collapsing them into two new columns: one for the names of the original columns and one for their values.

Let's create the wide-format tibble from our example and then tidy it.

```r
library(tidyverse)

# Create a "wide" tibble
cw_visits_wide <- tribble(
  ~site, ~"2023", ~"2024", ~"2025",
  "Governor's Palace", 305000, 312000, 315000,
  "Capitol Building",    280000, 295000, 302000,
  "Art Museums",         150000, 165000, 180000
)

print(cw_visits_wide)
```

```
## # A tibble: 3 x 4
##   site              `2023` `2024` `2025`
##   <chr>              <dbl>  <dbl>  <dbl>
## 1 Governor's Palace 305000 312000 315000
## 2 Capitol Building  280000 295000 302000
## 3 Art Museums       150000 165000 180000
```

Now, let's use `pivot_longer()` to tidy it. We need to tell the function: * `cols`: Which columns to pivot. Here, it's everything *except* the `site` column. * `names_to`: The name of the new column that will contain the former column headers (in our case, `"year"`). * `values_to`: The name of the new column that will contain the values that were in the pivoted columns (in our case, `"visits"`).

```
cw_visits_long <- cw_visits_wide %>%
  pivot_longer(
    cols = -site,
    names_to = "year",
    values_to = "visits"
  )

print(cw_visits_long)
```

```
## # A tibble: 9 x 3
##   site              year  visits
##   <chr>             <chr>  <dbl>
## 1 Governor's Palace 2023  305000
## 2 Governor's Palace 2024  312000
## 3 Governor's Palace 2025  315000
## 4 Capitol Building  2023  280000
## 5 Capitol Building  2024  295000
## 6 Capitol Building  2025  302000
## 7 Art Museums       2023  150000
## 8 Art Museums       2024  165000
## 9 Art Museums       2025  180000
```

Notice that the new `year` column is a character string. We can easily fix this by telling `pivot_longer` to parse the new column as a number.

```
# A more advanced pivot with type conversion
cw_visits_long_clean <- cw_visits_wide %>%
  pivot_longer(
    cols = -site,
    names_to = "year",
    values_to = "visits",
    names_transform = as.integer # Convert the new 'year' column to an integer
  )

glimpse(cw_visits_long_clean)
```

```
## Rows: 9
## Columns: 3
## $ site   <chr> "Governor's Palace", "Governor's Palace", "Governor's Palace
## $ year   <int> 2023, 2024, 2025, 2023, 2024, 2025, 2023, 2024, 2025
## $ visits <dbl> 305000, 312000, 315000, 280000, 295000, 302000, 150000, 165
```

This tidy format is now much easier to work with. For example, plotting visits over time is now trivial.

From Long to Wide: `pivot_wider()`

Sometimes, you need to do the reverse operation. `pivot_wider()` is the inverse of `pivot_longer()`. While you'll use it less often for analysis, it can be very useful for creating summary tables for reports.

Let's use the `cw_visits_long` tibble we just created to turn it back into its original wide format. We need to tell the function: * `names_from`: The column that contains the values that will become the new column *headers* (in our case, `year`). * `values_from`: The column that contains the values that will fill the cells of the new columns (in our case, `visits`).

```
cw_visits_long %>%
  pivot_wider(
    names_from = year,
    values_from = visits
  )
```

```
## # A tibble: 3 x 4
##   site              `2023` `2024` `2025`
##   <chr>              <dbl>  <dbl>  <dbl>
## 1 Governor's Palace 305000 312000 315000
## 2 Capitol Building  280000 295000 302000
## 3 Art Museums       150000 165000 180000
```

As you can see, we have perfectly reconstructed the original wide-format table. Understanding how to move data between these two formats is a critical skill for effective data wrangling.

Advanced Reshaping: Handling Complex Layouts

While simple pivots are common, real-world data is often more complex. You might have multiple identifier columns that need to stay fixed, and the column headers themselves might contain several pieces of information that you need to separate.

For this example, let's imagine we have water quality data from two monitoring sites on local rivers, "Jamestown" on the James River and "Yorktown" on the York River. For each site and date, both the mean and standard deviation (`sd`) were recorded for two different pollutants.

This "wide" format is very common for reporting experimental results:

```
library(tidyverse)

# A more complex "wide" tibble
water_quality_wide <- tribble(
  ~site,       ~collection_date, ~nitrogen_mean, ~nitrogen_sd, ~phosphorus_mea
  "Jamestown", "2025-10-01",     1.25,           0.15,         0.08,
  "Yorktown",  "2025-10-01",     0.95,           0.11,         0.06,
  "Jamestown", "2025-10-02",     1.31,           0.12,         0.09,
  "Yorktown",  "2025-10-02",     0.99,           0.09,         0.07,
)

print(water_quality_wide)
```

```
## # A tibble: 4 x 6
```

```
##   site   collection_date nitrogen_mean nitrogen_sd phosphorus_mean phospho
##   <chr>  <chr>                   <dbl>       <dbl>           <dbl>
## 1 James~ 2025-10-01               1.25        0.15            0.08
## 2 Yorkt~ 2025-10-01               0.95        0.11            0.06
## 3 James~ 2025-10-02               1.31        0.12            0.09
## 4 Yorkt~ 2025-10-02               0.99        0.09            0.07
```

Our goal is to tidy this data. A tidy version would have columns for `site`, `collection_date`, `pollutant`, `mean`, and `sd`.

More Complex `pivot_longer()`

The problem here is that the column names like `nitrogen_mean` contain two pieces of information: the pollutant ("nitrogen") and the statistic measured ("mean"). `pivot_longer()` can handle this elegantly using a special name, `.value`.

By including `.value` in the `names_to` argument, you tell `tidyr` that part of the column name should be treated as the *name* for a new column that will contain the cell values.

- `cols`: We select all columns that contain an underscore _.
- `names_to`: We provide a vector, `c("pollutant", ".value")`. This tells `pivot_longer` to split the old column names into two parts. The part before the separator will go into a new column called `pollutant`, and the part after the separator (which corresponds to `.value`) will become the names of the new *value* columns (in this case, `mean` and `sd`).
- `names_sep`: We specify the character to split the names by, which is _.

```r
water_quality_long <- water_quality_wide %>%
  pivot_longer(
    cols = contains(c("nitrogen_","phosphorus_")),
    names_to = c("pollutant", ".value"),
    names_sep = "_"
  )

# Let's print the result to confirm the column names
print(water_quality_long)
```

```
## # A tibble: 8 x 5
##   site      collection_date pollutant   mean    sd
##   <chr>     <chr>           <chr>      <dbl> <dbl>
## 1 Jamestown 2025-10-01      nitrogen    1.25  0.15
## 2 Jamestown 2025-10-01      phosphorus  0.08  0.02
## 3 Yorktown  2025-10-01      nitrogen    0.95  0.11
## 4 Yorktown  2025-10-01      phosphorus  0.06  0.01
## 5 Jamestown 2025-10-02      nitrogen    1.31  0.12
## 6 Jamestown 2025-10-02      phosphorus  0.09  0.02
## 7 Yorktown  2025-10-02      nitrogen    0.99  0.09
## 8 Yorktown  2025-10-02      phosphorus  0.07  0.01
```

This single, powerful command has perfectly reshaped our complex wide data into a tidy format. We have our two identifier variables (`site` and `collection_date`) preserved, and we've successfully created new columns for the pollutant and its corresponding mean and standard deviation.

More Complex `pivot_wider()`

Now, let's perform the inverse operation to demonstrate a more complex `pivot_wider()` call. Our goal is to take the tidy `water_quality_long` tibble and reshape it back to its original complex wide format.

Here, we have multiple value columns (`mean` and `sd`) that we want to spread out. `pivot_wider()` can handle this by accepting a vector for the `values_from` argument. It will automatically combine the names from `names_from` and `values_from` to create the new column headers.

- `id_cols`: We can explicitly state which columns are identifiers and should not be pivoted.
- `names_from`: The column whose values will become *part* of the new column names (in our case, `pollutant`).
- `values_from`: The columns whose values will fill the cells of the new columns. We provide a vector: `c(mean, sd)`.

```
water_quality_long %>%
  pivot_wider(
    # The id_cols must match the exact column names from the input tibble.
    id_cols = c(site, collection_date),
    names_from = pollutant,
    values_from = c(mean, sd)
  )
```

```
## # A tibble: 4 x 6
##   site   collection_date mean_nitrogen mean_phosphorus sd_nitrogen sd_phosp
##   <chr>  <chr>                   <dbl>           <dbl>       <dbl>
## 1 James~ 2025-10-01               1.25            0.08        0.15
## 2 Yorkt~ 2025-10-01               0.95            0.06        0.11
## 3 James~ 2025-10-02               1.31            0.09        0.12
## 4 Yorkt~ 2025-10-02               0.99            0.07        0.09
```

The result is identical to our original `water_quality_wide` tibble. The `pivot_wider()` function automatically created the combined column names (e.g., `mean_nitrogen`, `sd_nitrogen`) for us. These advanced features of `tidyr` provide the flexibility to handle nearly any data reshaping task you might encounter.

Case Study: From Messy Report to Final Plot

In this section, we will bring together everything we have learned. Data wrangling is not about applying functions in isolation; it's about building a sequential pipeline of operations that turns messy, raw data into a clean, tidy dataset ready for analysis and visualization.

Our Scenario: Imagine we are environmental analysts based in Williamsburg, Virginia, and our team is preparing to launch a new, local air quality monitoring project. Before we begin collecting our own data, our task is to develop and validate a data wrangling workflow in R. To do this, we will use a famous, publicly available historical dataset as a stand-in: the built-in `airquality` dataset, which contains records from New York City in 1973.

Our goal is to create a reproducible script that takes a "messy report" based on this historical data, tidies it completely, and produces a summary plot. This validated script can then be easily adapted for our real

Williamsburg data once it becomes available.

Step 1: Assessing the "Messy Report"

First, let's load the Tidyverse. Then, to simulate the kind of messy, wide-format report we might receive from a monitoring station, we will programmatically reformat the built-in `airquality` data.

```
library(tidyverse)

# --- This code simulates the messy report based on the NYC data ---
messy_report <- airquality %>%
  # We're interested in the summer months for our pilot study
  filter(Month %in% c(5, 6, 7)) %>%
  # Select a few relevant columns
  select(Day, Month, Wind, Temp) %>%
  # Pivot the data so each month's temperature is a different column
  pivot_wider(names_from = Month, values_from = Temp, names_prefix = "Temp_Mo

# Now, let's look at the "report" we need to clean
print(messy_report)
```

```
## # A tibble: 89 x 5
##      Day  Wind Temp_Month_5 Temp_Month_6 Temp_Month_7
##    <int> <dbl>        <int>        <int>        <int>
##  1     1   7.4           67           NA           NA
##  2     2   8             72           NA           NA
##  3     3  12.6           74           NA           NA
##  4     4  11.5           62           NA           NA
##  5     5  14.3           56           NA           NA
##  6     6  14.9           66           NA           NA
##  7     7   8.6           65           NA           NA
##  8     8  13.8           59           NA           NA
##  9     9  20.1           61           NA           NA
## 10    10   8.6           69           NA           NA
## # i 79 more rows
```

Let's use `glimpse()` to diagnose the problems:

```
glimpse(messy_report)
```

```
## Rows: 89
## Columns: 5
## $ Day          <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16
## $ Wind         <dbl> 7.4, 8.0, 12.6, 11.5, 14.3, 14.9, 8.6, 13.8, 20.1, 8.6
## $ Temp_Month_5 <int> 67, 72, 74, 62, 56, 66, 65, 59, 61, 69, 74, 69, 66, 68
## $ Temp_Month_6 <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
## $ Temp_Month_7 <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
```

Diagnosis: This report format is not suitable for analysis. We can identify several key problems: 1.

It's "Wide" Data: The columns `Temp_Month_5`, `Temp_Month_6`, and `Temp_Month_7` are not variables; they are *values* of a "month" variable. 2. **Redundant Rows and NAs:** The wide format creates rows with many NA (missing) values, as a given day only exists in one month. 3. **Clunky Column Names:** The column names are not ideal for coding.

Step 2: Reshaping with `pivot_longer()`

Our first priority is to fix the fundamental structural problem by converting the data from a wide to a long format. We will use `pivot_longer()` to collapse the temperature columns into a "month" column and a "temperature" column.

```
airquality_long <- messy_report %>%
  pivot_longer(
    cols = starts_with("Temp_Month_"), # Select all columns that start with "
    names_to = "Month",                # The new column for the old column nam
    values_to = "Temperature"          # The new column for the cell values
  )

# Let's see the result
head(airquality_long, 10)
```

```
## # A tibble: 10 x 4
##      Day  Wind Month        Temperature
##    <int> <dbl> <chr>              <int>
##  1     1   7.4 Temp_Month_5          67
##  2     1   7.4 Temp_Month_6          NA
##  3     1   7.4 Temp_Month_7          NA
##  4     2   8   Temp_Month_5          72
##  5     2   8   Temp_Month_6          NA
##  6     2   8   Temp_Month_7          NA
##  7     3  12.6 Temp_Month_5          74
##  8     3  12.6 Temp_Month_6          NA
##  9     3  12.6 Temp_Month_7          NA
## 10     4  11.5 Temp_Month_5          62
```

This is a huge improvement! Our data is now in a much longer, tidier format, which will be easier to clean and analyze.

Step 3: Cleaning and Transforming with `dplyr`

Now that the data has the correct shape, we can build a `dplyr` pipeline to finish the cleaning process.

```
airquality_tidy <- airquality_long %>%
  # 1. Remove rows where the Temperature is NA. These were artifacts of the p
  filter(!is.na(Temperature)) %>%

  # 2. Clean the 'Month' column by removing the prefix
```

```r
  mutate(Month = str_replace(Month, "Temp_Month_", "")) %>%

  # 3. Rename columns to be cleaner and more consistent (snake_case)
  rename(
    day = Day,
    wind_speed = Wind,
    temperature_F = Temperature,
    month = Month
  ) %>%

  # 4. Convert month from a character to a number for easier sorting later
  mutate(month = as.integer(month)) %>%

  # 5. Arrange the data logically
  arrange(month, day)

# Inspect our final, tidy tibble
glimpse(airquality_tidy)
```

```
## Rows: 92
## Columns: 4
## $ day           <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16
## $ wind_speed    <dbl> 7.4, 8.0, 12.6, 11.5, 14.3, 14.9, 8.6, 13.8, 20.1, 8
## $ month         <int> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,
## $ temperature_F <int> 67, 72, 74, 62, 56, 66, 65, 59, 61, 69, 74, 69, 66, 6
```

This is our final, analysis-ready dataset. Every column is a variable, every row is a unique observation, and every cell has a value.

Step 4: Summarizing the Data

With our clean data, the final analysis step is easy. We want to find the average wind speed for each month to test our summary logic.

```r
monthly_summary <- airquality_tidy %>%
  group_by(month) %>%
  summarise(
    avg_wind_speed = mean(wind_speed)
  ) %>%
  ungroup()

print(monthly_summary)
```

```
## # A tibble: 3 x 2
##   month avg_wind_speed
##   <int>          <dbl>
## 1     5           11.6
```

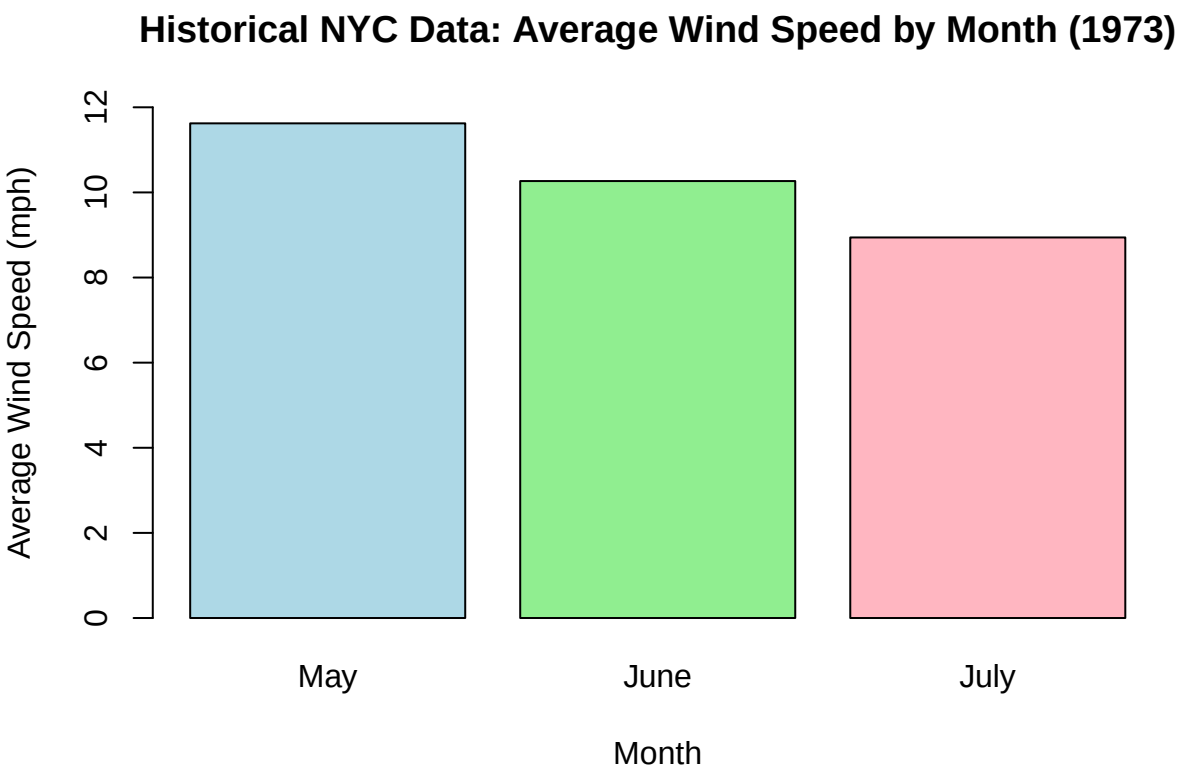
```
## 2      6            10.3
## 3      7             8.94
```

This summary tibble contains the exact information we need to create our final plot.

Step 5: The Payoff – The Final Plot

Now we will visualize our summary. The title and labels will be precise, clearly stating the source and content of the data, which is a critical part of any good analysis.

```r
# We use the barplot() function from base R
barplot(
  height = monthly_summary$avg_wind_speed,
  names.arg = c("May", "June", "July"),
  main = "Historical NYC Data: Average Wind Speed by Month (1973)",
  xlab = "Month",
  ylab = "Average Wind Speed (mph)",
  col = c("lightblue", "lightgreen", "lightpink"),
  ylim = c(0, 12)
)
```



Conclusion

We have successfully completed our goal. We started with a messy, report-style dataset and built a reproducible R script that transformed it into a clean, tidy format. From this tidy data, we calculated a summary statistic and created a clear visualization.

The key takeaway is that we now have a validated workflow. When we receive our real air quality data from Williamsburg, we can confidently adapt this script to quickly produce similar analyses. This is the power of a well-designed data wrangling pipeline.

A Quick Tour of Base R Plots

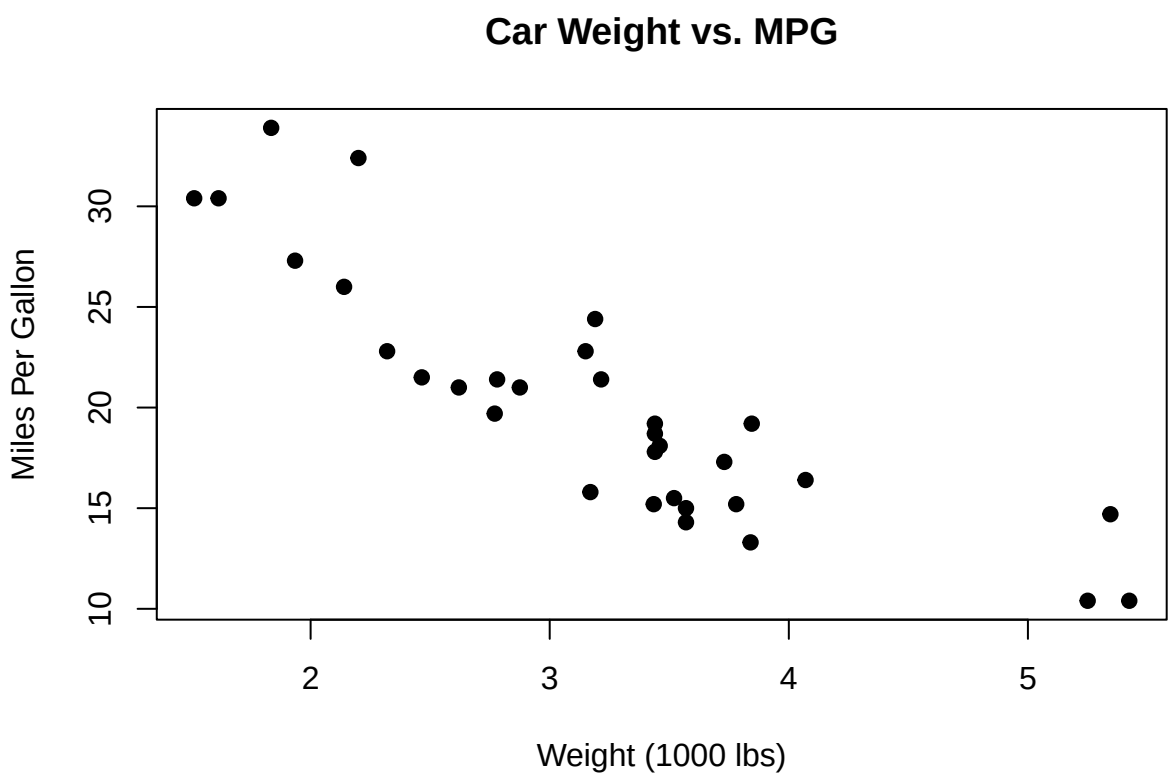
While ggplot2 is the modern standard, R's built-in plotting functions are fast and great for quick exploratory work without loading any additional libraries.

Base R Plot: Scatterplot (plot)

Purpose: To visualize the relationship between two continuous variables.

Example: Plot car weight vs. miles per gallon from the built-in 'mtcars' dataset

```
plot(mtcars$wt, mtcars$mpg,
     main = "Car Weight vs. MPG",
     xlab = "Weight (1000 lbs)",
     ylab = "Miles Per Gallon",
     pch = 19) # pch changes the point shape
```

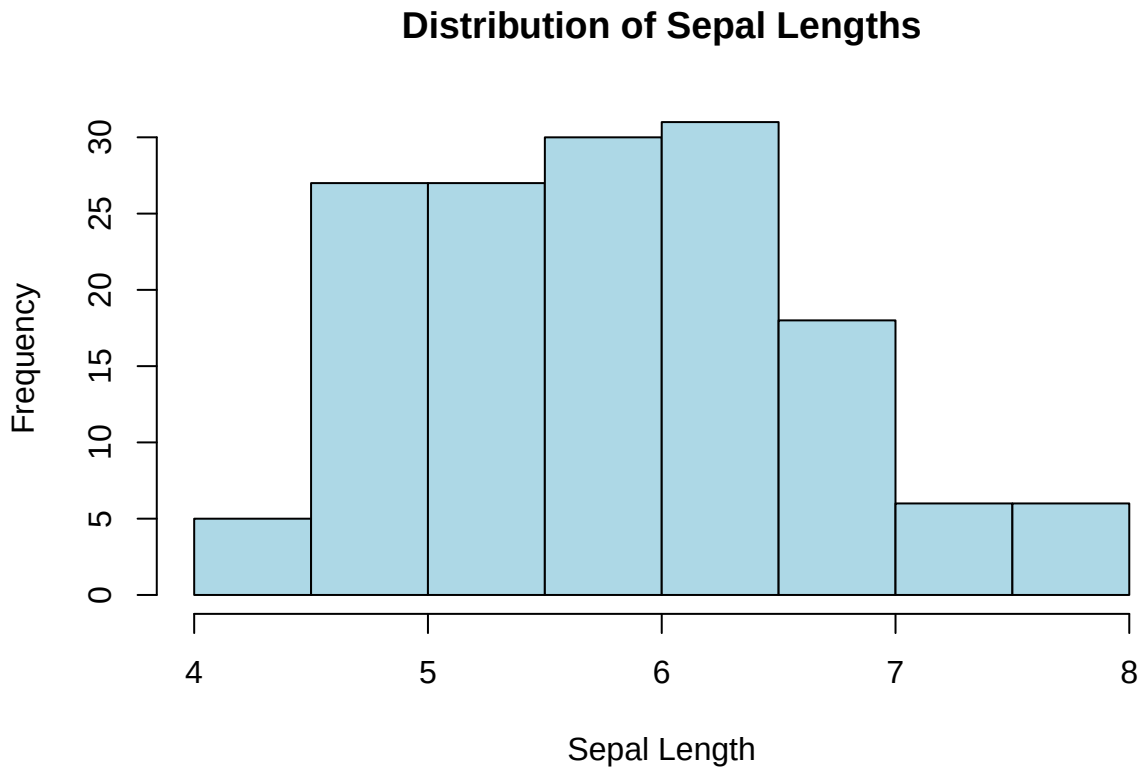


Base R Plot: Histogram (hist)

Purpose: To understand the distribution of a single continuous variable.

Example: Plot the distribution of sepal lengths from the built-in 'iris' dataset

```r
hist(iris$Sepal.Length,
     main = "Distribution of Sepal Lengths",
     xlab = "Sepal Length",
     col = "lightblue") # col sets the color
```

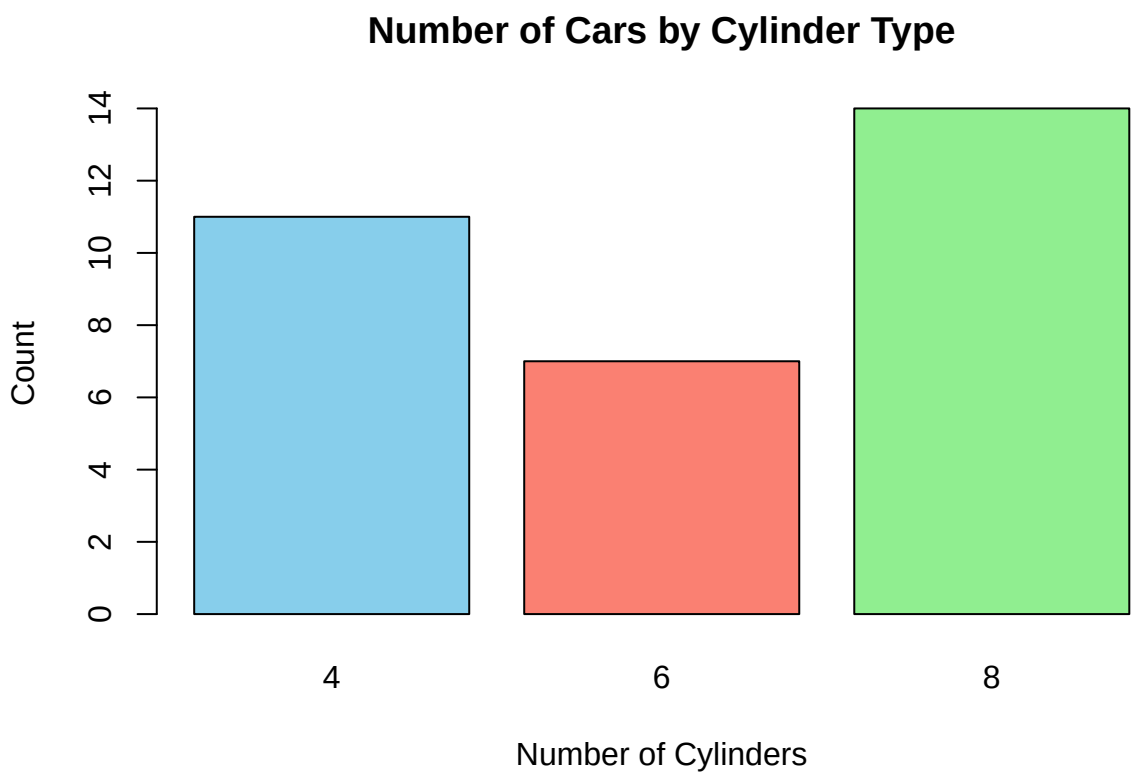


Base R Plot: Bar Plot (barplot)

Purpose: To compare the size of different categories. Note: barplot usually needs pre-summarized data (like averages across conditions).

```r
# First, create a summary table of counts for each cylinder type
cylinder_counts <- table(mtcars$cyl)

# Now, create the bar plot of the counts
barplot(cylinder_counts,
        main = "Number of Cars by Cylinder Type",
        xlab = "Number of Cylinders",
        ylab = "Count",
        col = c("skyblue", "salmon", "lightgreen"))
```

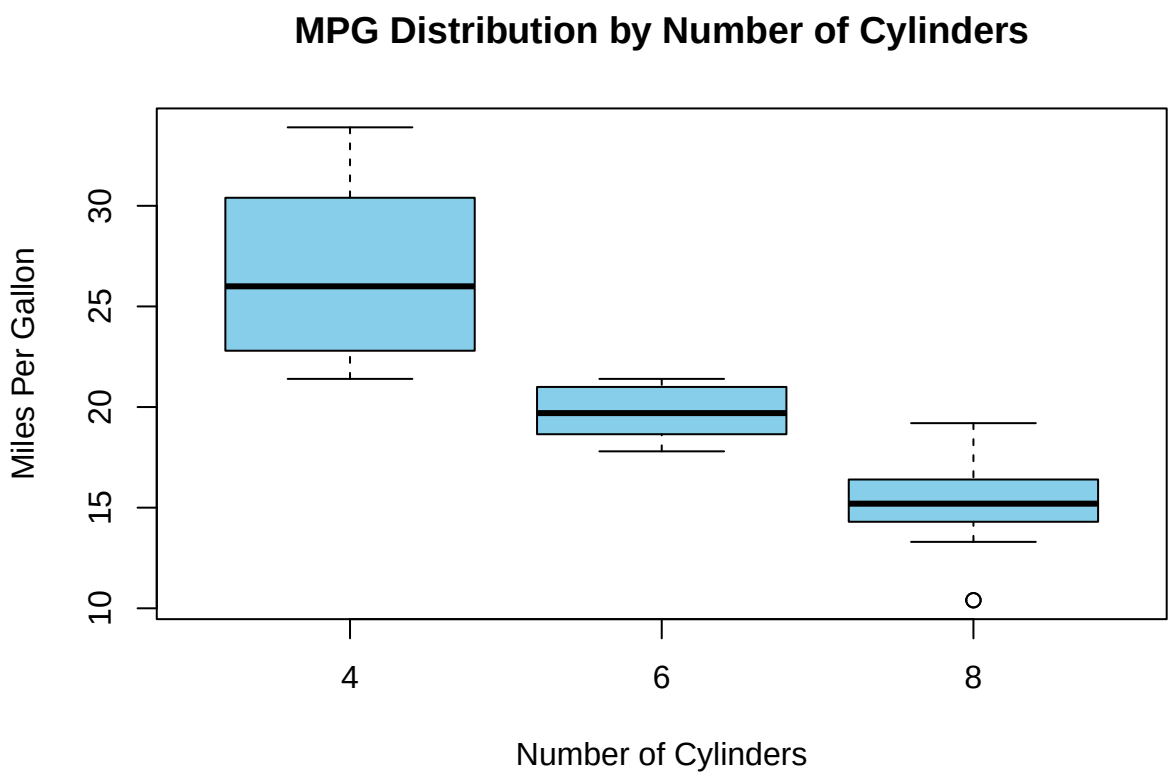


Base R Plot: Box Plot (boxplot)

Purpose: To compare the distribution of a continuous variable across several groups.

Example: Compare MPG distribution for each cylinder type using formula notation

```
boxplot(mpg ~ cyl,
        data = mtcars,
        main = "MPG Distribution by Number of Cylinders",
        xlab = "Number of Cylinders",
        ylab = "Miles Per Gallon",
        col = "skyblue")
```



Exercises

The following exercises will test your ability to apply the data wrangling skills you've learned in this chapter. Each problem uses datasets that are built into the Tidyverse, so you will need to have the `tidyverse` library loaded. The goal is not just to get the right answer, but to write a clean, readable pipeline of commands for each task. **Don't forget to use text blocks and/or comments to document your thinking and explain what each line of your code is doing.**

```
library(tidyverse)
```

Problem 1: Analyzing US Arrests

Dataset: `USArrests` (built into R)

Scenario: A law student at William & Mary is researching historical crime data from 1973 and wants to identify states with high urban populations and their corresponding violent crime rates.

Task: Write a single pipeline of `dplyr` commands that: 1. Starts with the `USArrests` dataset. Note that the state names are currently row names. You can turn them into a proper column by running `USArrests %>% rownames_to_column(var = "state")` first. 2. Filters the data to include only states with an `UrbanPop` rate of 80% or higher. 3. Selects only the `state`, `Murder`, and `Rape` columns. 4. Arranges the final result by the `Murder` rate in descending order. 5. Write a sentence or two to describe the result.

```
# Your code here
```

Problem 2: Mammal Sleep Habits

Dataset: `msleep` (from the `ggplot2` package)

Scenario: A researcher at the Virginia Living Museum wants to know if there's a relationship between an animal's diet and its sleep patterns, specifically the proportion of time spent in REM sleep.

Task: Write a `dplyr` pipeline that: 1. Starts with the `msleep` dataset. 2. Removes rows where the `vore` (diet) column is NA, as we can't use them for this analysis. (Hint: use `filter(!is.na(vore))`). 3. Creates a new column called `rem_proportion` which is the ratio of `sleep_rem` to `sleep_total`. 4. Groups the data by `vore`. 5. Calculates the average `rem_proportion` for each diet group. 6. Arranges the final tibble to show the diets with the highest average REM proportion first. 7. Create a bar chart or boxplot to visualize the REM proportion for each diet type. 8. Write a sentence or two to describe the results.

```
# Your code here
```

Problem 3: Tidying Global Health Data

Dataset: `who` (from the `tidyr` package)

Scenario: An intern at the Virginia Department of Health has been given a dataset from the World Health Organization about tuberculosis cases. The data is in an extremely "wide" and messy format and needs to be tidied before any analysis can be done.

Task: This is a classic reshaping challenge. The columns from `new_sp_m014` through `newrel_f65` contain values that should be in their own columns. 1. Start with the `who` dataset. 2. Use `pivot_longer()` to gather all the columns from `new_sp_m014` to `newrel_f65`. 3. The names of these columns should go into a new column called `diagnosis_code`. 4. The values should go into a new column called `count`. 5. After pivoting, many rows will have a `count` of NA. Filter these rows out. 6. Display the first few rows of your final tidy tibble.

(Hint: This is a complex pivot! You can select the columns with `cols = new_sp_m014:newrel_f65`)

```
# Your code here
```

Problem 4: Finding the Windiest Year for Storms

Dataset: `storms` (from the `dplyr` package)

Scenario: Meteorologists in Hampton Roads are analyzing the historical Atlantic storm dataset to identify the year with the highest average maximum wind speeds, which could indicate a particularly active storm season.

Task: Write a `dplyr` pipeline to determine which year had the highest average wind speeds for its storms. Your final output should be a tibble with only one row (the winning year). 1. Start with the `storms` dataset. 2. Group the data by `year`. 3. For each year, calculate the average maximum wind speed (`wind`). 4. Arrange the results to find the year with the highest average. 5. Create a scatter plot of year as a function

of wind speed. 6. Return only the top row. (Hint: `head(1)` or `slice_max()` are good ways to do this). 7. Write a sentence or two to describe your results.

```
# Your code here
```

Problem 5: From Tidy back to Wide

Dataset: Your tidied data from Problem 3.

Scenario: After you tidied the `who` dataset, your manager asks for a specific summary table: a report showing the total number of cases for Afghanistan, Brazil, and China for the years 1999 through 2001. They want the years as columns for a quick comparison.

Task: This requires you to filter, summarize, and then reshape your tidy data back into a wide format. 1. Start with the tidied data you created in Problem 3. 2. `filter()` for only the countries of interest (`Afghanistan`, `Brazil`, `China`) and the years of interest (`1999`, `2000`, `2001`). 3. Calculate the total number of cases (`count`) for each country and year combination. (Hint: `group_by()` then `summarise()`). 4. Use `pivot_wider()` to reshape this summary table. The new column headers should come from the `year` column, and the values should come from your new `total_cases` column. 5. Write a sentence or two to describe your results.

```
# First, re-create your tidy 'who' data from problem 3 here
# tidy_who <- who %>% ...

# Then, write the pipeline for this problem
# tidy_who %>% ...
```

Problem 6: Ranking Species by Average BMI (Star Wars)

Dataset: starwars (from dplyr)

Scenario: A sci-fi physiology class wants to compare species by average Body Mass Index (BMI), computed from height and mass.

Task:

Write a pipeline that:

1. Starts from starwars, and keeps name, species, height, mass.
2. Creates height_m = height / 100 and bmi = mass / (height_m^2).
3. Groups by species, computes avg_bmi and n (characters per species).
4. Ungroup and return the top 5 species by avg_bmi, breaking ties by n (larger first).
5. Create a visualization (plot) of your results.
6. Write a sentence or two to describe your results.

(Tip: This reinforces mutate(), summarise(), arrange(), and the importance of ungroup().)

```
# Your code here
```

Problem 7: Advanced Reshaping with `.value` (Water Quality)

Dataset: Recreate the `water_quality_wide` tibble from the chapter (or copy it in your chunk).

Scenario: You need a tidy table that separates pollutant and statistic, then a compact report.

Task:
1. Start from `water_quality_wide` (columns like `nitrogen_mean`, `nitrogen_sd`, etc.).

2. Use `pivot_longer()` with `names_to = c("pollutant", ".value")` and `names_sep = "_"` to get columns `site, collection_date, pollutant, mean, sd`.

3. Compute **overall means by pollutant** (average of the `mean` column across sites/dates).

4. Return a **one-row per pollutant** report and then `pivot_wider()` so pollutants are columns and the overall means are values.

5. Write a sentence or two describing your results.

(Tip: This exercises the advanced `.value` pattern and a round-trip pivot.)

```
# Your code here
```